5. Scenario-Based Robustness and Sustainability Assessment

5.1 Robustness Assessment Framework

SWITCH must assess whether routing benefits survive changes in workload and prediction reliability. The assessment retains Task 2's frozen Pre-LightGBM direct router with its finite-sample risk guard and Task 3's final Spatial MILP + DES. Four scenarios compare the baseline of 7,335 requests with a 40% Math/Code share at unchanged volume, prediction error, and approximately 50% demand growth to 11,002 requests. Prediction error adds independent $\mathcal{N}(0, 0.10^2)$ perturbations to the two saved quality predictions before clipping to $[0, 1]$, then reapplies the frozen routing rule; duplicates share prompt-group perturbations. Math, Code and unregistered-source fallbacks remain L.

Each scenario uses three paired Greedy/Spatial seeds with identical routing, requests, six hosts and exogenous randomness: 24 executions over $H = 43{,}200$ s. Scheduling retains its causal information boundary; historical quality and future actual carbon are evaluation data. Summaries average seed-level quantities, including ratios and completed-request P95s. Three seeds support descriptive comparisons; replayed requests add no independent quality observations.

For historical score q_{i,m_i} and Task 3's violation indicator V_i, define

$$\begin{aligned} Q &= \frac{1}{N}\sum_i q_{i,m_i}, & P_{\text{hist}} &= \frac{1}{N}\sum_i \mathbf{1}\{q_{i,m_i} \geq 0.5\}, \\ P_{\text{valid}} &= \frac{1}{N}\sum_i (1 - V_i)\mathbf{1}\{q_{i,m_i} \geq 0.5\}, & \text{Violation} &= \frac{1}{N}\sum_i V_i. \end{aligned} \tag{5.1}$$

Failure, excessive waiting, late completion and window censoring count as violations. Historical quality is distinct from compliant delivery; completed-request P95 accompanies all-input service risk.

5.2 Robustness Under Three Stress Scenarios

Higher Share of Math/Code Requests. Increasing their share raises Spatial's P95 from 7.827 to 11.967 s, while violations remain 0.0182%. Mean historical Score falls from 0.776142 to 0.748728, but historical pass rate rises from 83.817% to 84.872%. This is consistent with the baseline Math distribution: its mean Score is lower than General's (0.6624 versus 0.8125), but its threshold-pass rate is higher (91.99% versus 81.51%). Mean score and threshold probability need not move together. The scenario represents task composition and computational load, rather than greater intrinsic difficulty for every request.

Prediction Error. Per seed, 133-139 requests change model choice: L-to-S counts are 67/75/68 and S-to-L counts are 70/64/65, all in General. Comparing the same changed requests using their paired historical S/L scores identifies 19/22/18 declines of at least 0.10 Score, a descriptive reporting threshold. Among new S choices, 13/15/12 lose an answer that L would have passed. The small population-average change conceals these local harms: fewer than 2% of requests switch, approximately 71.63% of switches have identical S/L scores, and gains offset losses. The fallback prevents Math/Code switches here; no guard-removal ablation or renewed statistical coverage under noise is established.

Demand Growth. At 11,002 requests, Spatial's violations rise from 0.0182% to 0.1697% and P95 to 11.885 s; completion is 99.8303%. Growth Greedy has more violations (1.6270%) but lower completed-request P95 (6.998 s). Spatial therefore trades longer observed tail response for fewer violations. Different completed sets preclude interpreting P95 alone as overall service superiority.

5.3 Carbon Rebound and Total Environmental Impact

The unchanged facility boundary is

$$C_{\text{total}} = C_{\text{idle}} + C_{\text{active}} + C_{\text{net}} + C_{\text{ctrl}}, \qquad C_{\text{op}} = C_{\text{total}} - C_{\text{idle}}. \tag{5.2}$$

The active term accounts for electricity above idle; network and controller costs remain separate. All powered hosts retain their full-window idle charge. Baseline Greedy emits 41,288.178 gCO2e and Spatial 41,214.556 gCO2e: a same-demand saving of 73.622 gCO2e, or 0.1783%. Fixed idle contributes 41,140.3512 gCO2e, approximately 99.82% of baseline Spatial's total. This is an accounting share, not idle-time utilization or a removable energy budget. Spatial uses approximately 0.004857 kWh more electricity while emitting less carbon: carbon efficiency is not necessarily equivalent to energy efficiency.

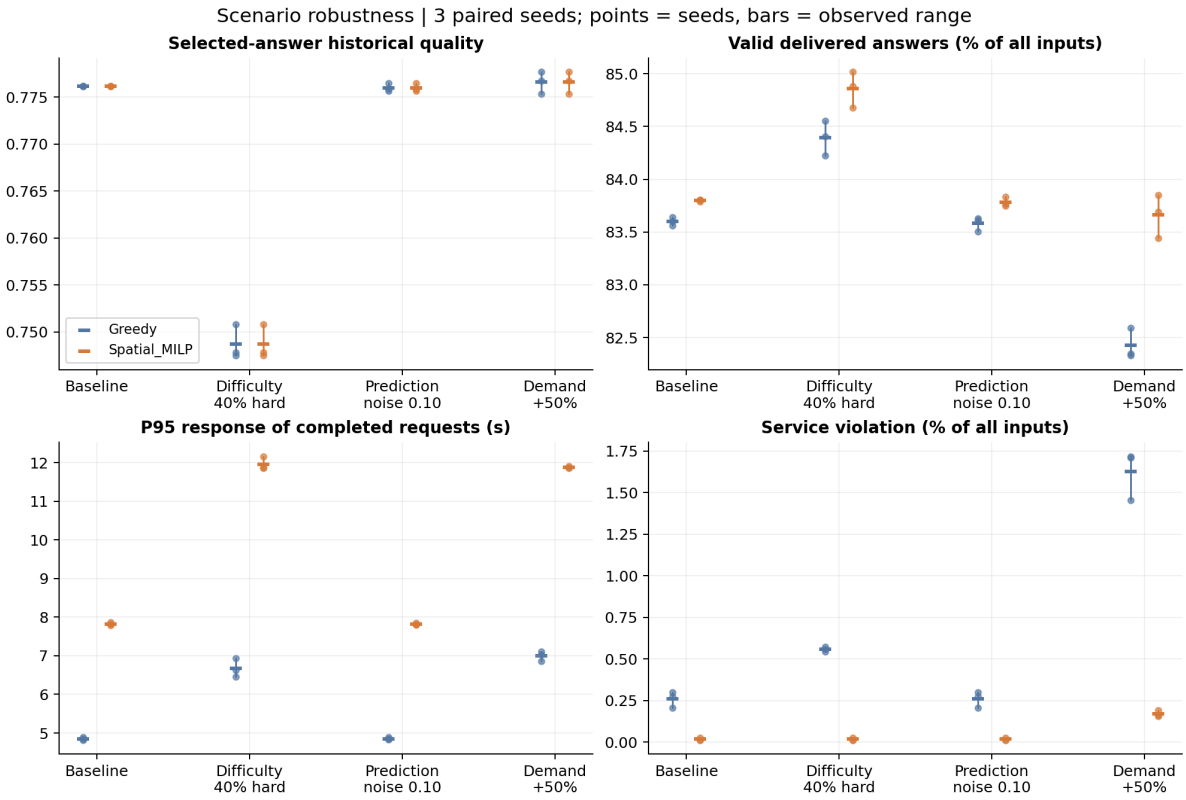


Figure 1: Paired scenario results. Points denote seeds, horizontal marks means, and vertical bars observed ranges. P95 is conditional on completion; violation and valid-pass rates use all inputs. The original label "Difficulty 40% hard" means a 40% Math/Code share.

Let $C_G(N)$ and $C_{\mathrm{Sp}}(N)$ denote total carbon under Greedy and Spatial at demand N. With $N_0 = 7{,}335$ and $N_1 = 11{,}002$, the paired mean contrasts are

$$\begin{aligned} C_G(N_1) - C_{\mathrm{Sp}}(N_1) &= 101.443 \text{ gCO2e}, \\ C_{\mathrm{Sp}}(N_1) - C_G(N_0) &= +2.496 \text{ gCO2e}. \end{aligned} \tag{5.3}$$

Spatial still saves carbon at matched increased demand, yet its growth total, 41,290.674 gCO2e, exceeds original-demand Greedy by approximately 0.006045%. These comparisons answer different questions. The small reversal is conditional on power, PUE and carbon-intensity assumptions; its real-world sign and magnitude are unverified.

Growth also changes where computation occurs. France's share of Spatial's active electricity falls from 89.583% to 79.632%, while execution-weighted active carbon intensity rises from 42.237 to 57.231 gCO2e/kWh. The regional hourly carbon series are unchanged: execution weights shift toward more carbon-intensive regions. Consequently the baseline per-input operational carbon coefficient is not constant under growth.

Under fixed idle and constant $c_0 = [C_{\mathrm{Sp}}(N_0) - C_{\mathrm{idle}}]/N_0 > 0$, the linear reference $C^{\mathrm{lin}}_{\mathrm{Sp}}(N) = C_{\mathrm{idle}} + Nc_0$ yields

$$r_{\mathrm{lin}} = \frac{C_G(N_0) - C_{\mathrm{Sp}}(N_0)}{C_{\mathrm{Sp}}(N_0) - C_{\mathrm{idle}}}. \tag{5.4}$$

Its seed-mean value, 99.21%, is algebraically correct under those assumptions, but is not a DES rebound threshold. Only zero and approximately 50% growth were simulated; an accurate nonlinear crossing is unidentified. Moreover, growth was imposed externally. Without demand-elasticity evidence, this comparison cannot establish that efficiency improvements caused increased use.

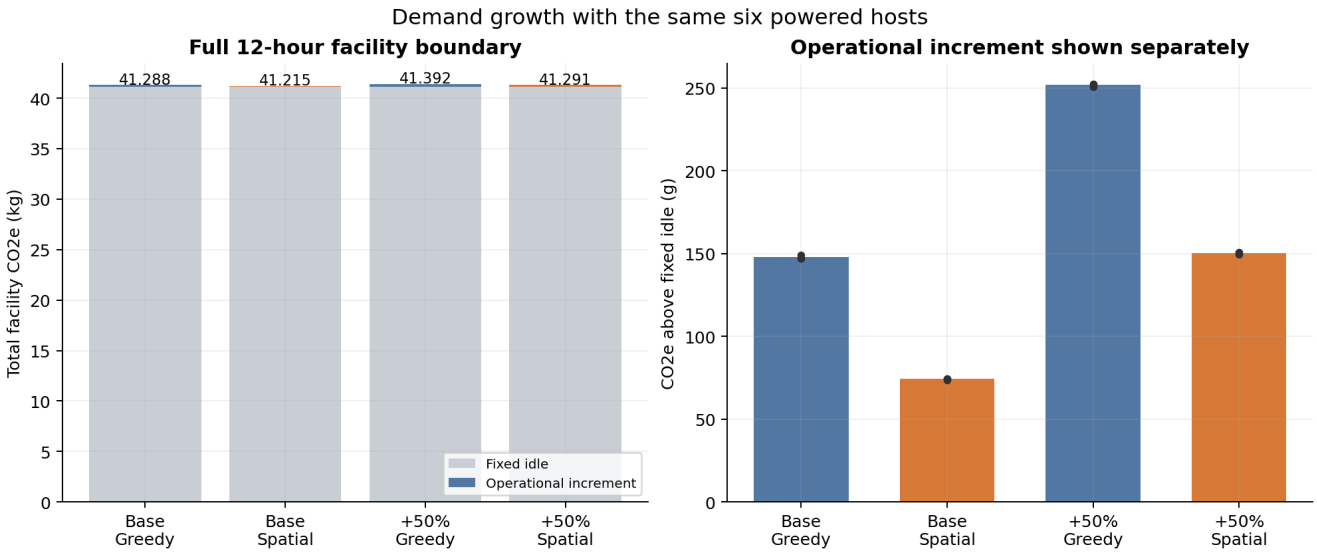


Figure 2: The left panel includes fixed idle and uses kgCO2e; the right isolates operational increments in gCO2e. Greedy versus Spatial within demand measures scheduling savings; growth Spatial versus baseline Greedy measures the cross-demand total change.

5.4 Fairness Across Request Types

Routing quality and scheduling service require separate comparisons. Using paired historical scores, the routing contrast is $\Delta Q_g = N_g^{-1} \sum_{i \in g} (q_{i,m_i} - q_{i,L})$. Baseline General has Router/All-L means 0.812466/0.818424, a difference of -0.005958 and a pass-rate difference of -0.5055 percentage points. Math/Code choose L throughout, giving zero reference differences, rather than perfect absolute quality. L does not dominate S on every request. All-L is solely an offline quality reference, with no simulated All-L service or carbon counterfactual.

Across the 36 task-by-scenario-by-seed comparisons, Spatial never increases observed category violations relative to Greedy, but completed-request P95 is generally higher. Table 1 shows growth outcomes. Code has the highest Spatial violation rate and largest increase from baseline, yet its 86 baseline instances contain only 85 distinct prompt groups; duplication adds no independent quality evidence. Task comparisons neither establish demographic fairness nor exclude other systematic harms.

Table 1. Growth scenario, seed means. Each cell is Greedy/Spatial. Rates use all inputs; P95 uses completions. Valid pass requires both historical quality and compliant delivery.

Task	Violation (%)	Completion (%)	P95 (s)	Valid pass (%)
General	1.5925/0.1404	98.408/99.860	4.896/10.502	80.247/81.438
Math	1.7354/0.2482	98.265/99.752	8.493/13.945	90.293/91.703
Code	1.7590/0.4919	98.241/99.508	12.211/18.329	67.435/68.204

5.5 Environmental Evidence and Limitations

Table 2 separates environmental observations, transferred parameters and assumptions from deployment evidence still needed.

Table 2. Evidence supporting the assessment and its limits.

Evidence used	What remains unestablished
RouterBench paired historical S/L scores	Representative public-sector quality and demand
Historical hourly DE/FR/ES carbon intensity	Original forecast vintages and marginal-grid effects
GPU power proxy transferred from published measurements	Actual power of the target models on their serving hardware
Assumed idle power, PUE, arrivals and service times	Site telemetry, calibrated queues and facility PUE
Conditional DES execution and complete energy ledger	Measured deployment savings and a suitable causal counterfactual

Historical average intensity supports attributional accounting, not marginal-grid causality. SWITCH should require measured site energy and PUE, representative quality and service audits, and monitoring of demand and task-level harms. Matched-demand carbon savings here assure neither lower electricity consumption, shorter responses, nor lower totals after growth.